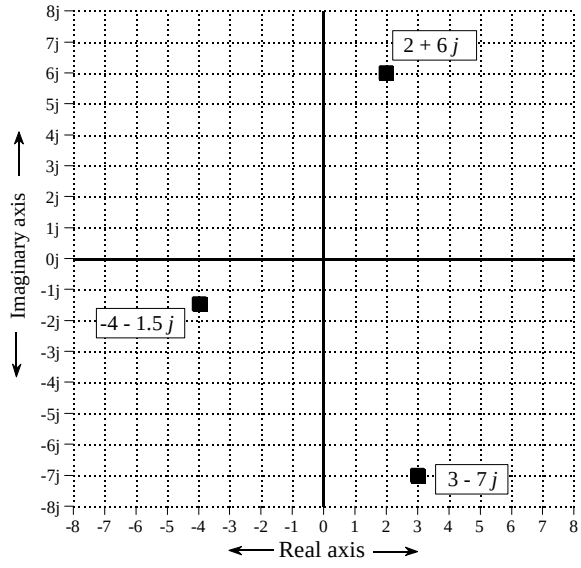

FIGURE 30-1
The complex plane. Every complex number has a unique location in the complex plane, as illustrated by the three examples shown here. The horizontal axis represents the real part, while the vertical axis represents the imaginary part.

complex plane is the real part of the complex number, while the vertical axis is the imaginary part. Since real numbers are those complex numbers that have an imaginary part equal to zero, the *real number line* is the same as the *x-axis* of the complex plane.

In mathematical equations, a complex number is represented by a single variable, even though it is composed of two parts. For example, the three complex variables in Fig. 30-1 could be written:

$$\begin{aligned} A &= 2 + 6j \\ B &= -4 - 1.5j \\ C &= 3 - 7j \end{aligned}$$

where A, B, & C are complex variables. This illustrates a strong *advantage* and a strong *disadvantage* of using complex numbers. The advantage is the inherent shorthand of representing two things by a single symbol. The disadvantage is having to remember which variables are complex and which variables are ordinary numbers.

The mathematical notation for separating a complex number into its real and imaginary parts uses the operators: *Re*( ) and *Im*( ). For example, using the above complex numbers:

$$\begin{aligned} Re\ A &= 2 & Im\ A &= 6 \\ Re\ B &= -4 & Im\ B &= -1.5 \\ Re\ C &= 3 & Im\ C &= -7 \end{aligned}$$